

SUMMARY

- IT professional with over 2 years of experience in analysis, design and development of scalable web applications using Java, Spring Boot, and React, deployed on AWS.
- Proficient in building cloud-native microservices and developing interactive frontends connected to backend services.
- Hands-on experience in orchestrating containerized application deployments using Kubernetes.
- Well-versed in Agile practices, test-driven development (TDD), and CI/CD pipelines using Jenkins.
- Strong foundation in system design and software development methodologies.
- Effective collaborator with strong communication and problem-solving skills.
- Worked on AI projects involving PyTorch models and Federated Learning, applying machine learning techniques to real-world problems.
- Familiar with data analysis and visualization tools such as Power BI and R programming.

EDUCATION

Master of Science in Computer Science , <i>University of Houston – Main Campus</i>	08/2023 – 05/2025
<i>Courses: Web Development, Software Design, Cloud Computing, Networking, Machine Learning, Artificial Intelligence.</i>	
Bachelor of Engineering in Computer Science , <i>National Institute of Technology (NIT) Durgapur</i>	08/2019 – 07/2023
<i>Courses: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Compiler Design.</i>	

TECHNICAL SKILLS

Programming & Scripting	Java, C/C++, Python, PyTorch, R, Bash, JavaScript, TypeScript, awk
Frontend Development	React, HTML5, Ajax, jQuery, Tailwind CSS, Bootstrap, Material UI
Backend Development	Spring, Spring Boot, Spring MVC, Kafka, Zookeeper, Node.js, REST, JWT, OAuth 2.0
Application Servers	Apache Tomcat, nginx
Testing Tools	JUnit, Mockito, Jest, WireMock, React Testing Library
Build & Automation	Maven, Gradle, Vite, Babel
Databases	Oracle, MySQL, Redis, DynamoDB
Cloud Platforms	AWS (EC2, S3, RDS, Lambda, CloudFront, Amplify, EKS)
DevOps & CI/CD	Git, SVN, Jenkins, Kubernetes, Docker, Postman, VS Code, Confluence
Monitoring & Observability	Grafana, Prometheus, Elasticsearch, Logstash, Kibana
Engineering Practices	Microservices Architecture, Agile (Scrum), CI/CD Pipelines, System Design, Code Reviews

EXPERIENCE

University of Houston – Main Campus, TX	01/2025 – 05/2025
CS6339 – BookBazaar: Campus Bookstore Website	

- Architected and developed a full-stack e-commerce web application simulating real-world transactions and business workflows.
- Implemented secure authentication and session management using **JWT** and **Spring Security**, ensuring data integrity and user role isolation.
- Designed and normalized a relational database in **MySQL** to support product catalog, shopping cart, inventory, and order management modules.
- Built scalable RESTful APIs using **Spring Boot**, applying layered architecture and best practices for code reusability and maintainability.
- Integrated **Stripe API** for handling secure payments, invoice generation, and refund processing, aligned with PCI compliance standards.
- Developed a responsive and interactive user interface with **React**, implementing conditional rendering, routing, and state management.
- Applied role-based access control to segregate customer and admin functionality, enhancing application security and usability.
- Conducted functional and integration testing using Postman and Swagger UI to validate endpoints and workflows.
- Deployed the application locally on **Apache Tomcat** and managed development using **Git**, **Maven**, and **VS Code**.

Environment: Java 11, Spring Boot, Spring Security, Tomcat, React, MySQL, Stripe API, JWT, Maven, Git, REST APIs, Visual Studio Code

University of Houston – Main Campus, TX	08/2024 – 11/2024
CS6368 – ModelServeX: Benchmarking TensorFlow vs TorchServe on AWS	

- Benchmarked ML inference performance of TensorFlow Serving and TorchServe on AWS EC2 with MobileNetV2 and Fashion-MNIST.
- Automated training, packaging, and container-based deployment workflows using Docker.
- Analyzed latency, throughput, and CPU/memory usage under varying traffic conditions, delivering visual performance reports.
- Recommended optimal deployment strategies based on load profiles and service trade-offs.

Environment: Python, C++, TensorFlow Serving, TorchServe, ONNX, MobileNetV2, Fashion-MNIST, Docker, AWS EC2, Shell, Bash, Jupyter

University of Houston – Main Campus, TX
CS6364– EduQuiz: Secure Student Quiz Grade Management System

01/2024 – 05/2024

- Engineered secure REST APIs using Spring Boot, OAuth2, and JWT, documented with Swagger for maintainability.
- Used Flyway with Spring Data JPA for seamless schema migrations in PostgreSQL.
- Built a dynamic frontend using React with Context API and Axios for efficient state and data flow.
- Dockerized and deployed application on AWS EC2, configured HTTPS via Let's Encrypt, and implemented CI/CD via GitHub Actions.

Environment: Java 11, Spring Boot, Tomcat, React, PostgreSQL, OAuth2, JWT, Flyway, Docker, AWS EC2, GitHub Actions, Swagger, Axios, VS Code

University of Houston – Main Campus, TX
CS6344 - DataPulse: Real-Time Admin Dashboard

08/2023 – 11/2023

- Developed an enterprise-grade admin dashboard enabling real-time analytics, user management, and system performance tracking.
- Leveraged Redux Toolkit Query for scalable state and API management, enhancing data efficiency and modularity.
- Designed responsive UIs using Material UI and visualized analytics via Nivo; integrated REST APIs built with Express.js and MongoDB.
- Automated CI/CD deployment using Render, enabling rapid iteration cycles with zero-downtime releases.

Environment: React, Redux Toolkit, React Router, Material UI, Nivo, JavaScript, Vite, GitHub, Node.js, Express.js, MongoDB, Render

National Institute of Technology Durgapur, India
Intern – Programmer

10/2022 – 05/2023

- Designed and developed secure, modular RESTful APIs using **Spring Boot**, following service-layer architecture and best coding practices.
- Implemented backend logic and integrated **MySQL** and **PostgreSQL** databases using **Spring Data JPA**, optimizing queries for performance and scalability.
- Built an authentication and authorization system using **OAuth2** and **JWT** with fine-grained, role-based access control.
- Developed a responsive frontend interface using **React**, focusing on accessibility, performance, and cross-device compatibility.
- Utilized **React Hooks**, **React Router**, and **Redux Toolkit** for effective state and navigation management.
- Implemented form workflows and client-side validation using **Formik** and **Yup**, enhancing user experience and input accuracy.
- Transformed UI designs from Figma into dynamic, component-based interfaces using modern JavaScript and styling techniques.
- Integrated REST APIs with **Axios**, managing asynchronous data flows and robust error handling.
- Deployed the application using **AWS S3** and **CloudFront**, with automated CI/CD pipelines configured via **GitHub Actions**.

Environment: Java 11, Spring Boot, Spring Data JPA, OAuth2, JWT, MySQL, PostgreSQL, React, React Hooks, Redux Toolkit, React Router, Formik, Yup, Axios, GitHub Actions, AWS S3, CloudFront, Figma

Indian Institute of Technology Kharagpur, India
Intern – Programmer

05/2022 – 08/2022

Crop Classification from Satellite Imagery

- Conducted research on crop identification using satellite imagery and computer vision techniques.
- Engineered domain-specific features including pixel intensities, NDVI/AVI indices, and spatial context for improved classification accuracy.
- Built an ensemble classification model using **CatBoost** and **LDA** with bagging and class weighting to address class imbalance and increase precision.
- Evaluated model performance using accuracy, confusion matrices, and cross-validation strategies for generalization.

Environment: Python, CatBoost, LDA, NumPy, scikit-learn, Jupyter Notebook, Computer Vision, Git, VS Code

Federated Learning for Digital Pathology Image Classification in Distributed Medical Systems

- Designed and trained a federated learning model for pathology image classification on the Colorectal Cancer Histology dataset using **PyTorch** and **Flower**, achieving 87% accuracy.
- Simulated decentralized client environments with secure communication protocols and integrated **differential privacy** mechanisms to protect patient data.
- Implemented federated learning algorithms including **FedAvg**, **FedNova**, and **FedProx**, analyzing their performance under both **IID** and **non-IID** data distributions.
- Automated model training, evaluation (accuracy, loss, confusion matrix), and visualizations using CNN-based architectures for interpretability and insight.

Environment: Python, PyTorch, Flower, CNN, NumPy, Matplotlib, Seaborn, argparse, Git, VS Code, Jupyter Notebook

CERTIFICATIONS

AWS Certified Solutions Architect – Associate, Amazon Web Services (AWS)

Skills: Designing distributed systems, high availability, fault tolerance, cost optimization, and cloud-native architecture best practices on AWS.